```python
import pandas as pd
import plotly.express as px
import plotly.io as pio
pio.renderers.default = 'browser'
df = pd.read_csv(r"C:\Users\SANDEEP MEDIDA\Desktop\data-viz-class-materia
df = df.rename(columns={
    'Entity': 'Country',
    'Annual CO₂ emissions': 'CO2'
})
print(df.head())
print(df.columns)
df_filtered = df[(df['Year'] >= 2000) & (df['Year'] <= 2022)]
top_countries = df_filtered.groupby('Country')['CO2'].sum().nlargest(5).i
df_top = df_filtered[df_filtered['Country'].isin(top_countries)]
fig = px.line(
    df_top,
    x='Year',
    y='CO2',
    color='Country',
    title="CO₂ Emissions Trend (Top 5 Countries, 2000–2022)"
)
fig.update_layout(
    plot_bgcolor='white',
    xaxis_title="Year",
    yaxis_title="CO₂ Emissions"
)
fig.show()
```

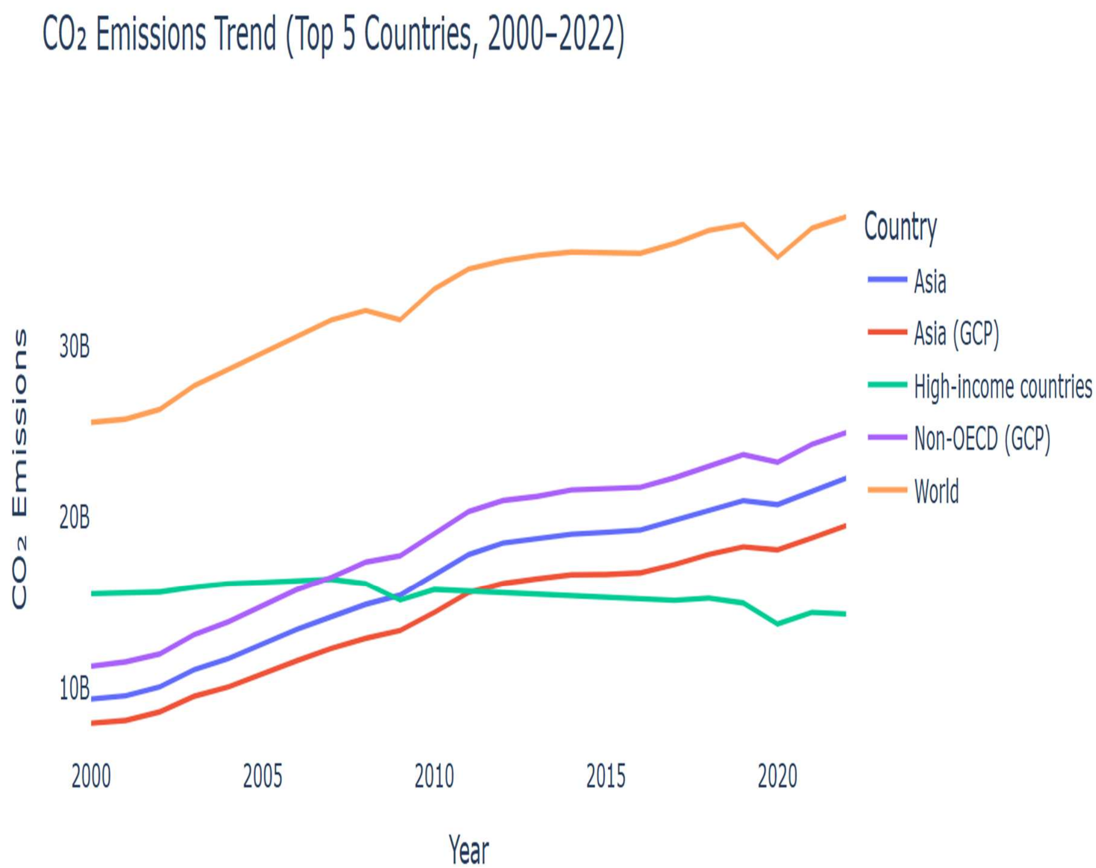
In [6]:

```
       Country Code  Year       CO2
0  Afghanistan  AFG  1949   14656.0
1  Afghanistan  AFG  1950   84272.0
2  Afghanistan  AFG  1951   91600.0
3  Afghanistan  AFG  1952   91600.0
4  Afghanistan  AFG  1953  106256.0
```

```
Index(['Country', 'Code', 'Year', 'CO2'],
dtype='object')
```

In [8]:

```python
print("Countries:", df['Country'].unique())

print("\nCO2 range:", df['CO2_Mt'].min(), "to", df['CO2_Mt'].max(), "Mt")

print("\nRegional averages (2022):")

print(
    df[df['Year'] == 2022]
     .groupby('Region')['CO2_Mt']
     .mean()
     .sort_values(ascending=False)
     .round(1)
)
```

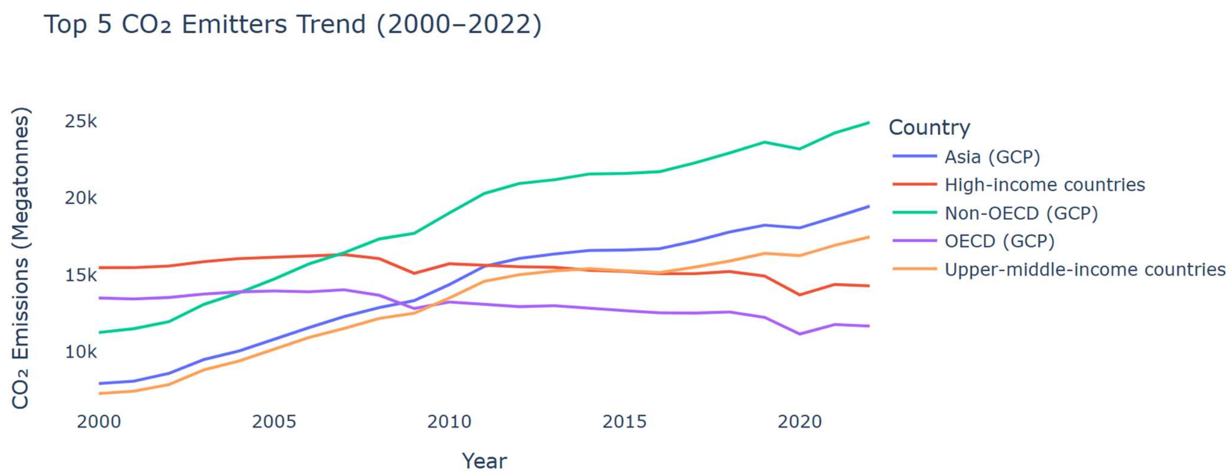

CO₂ Emissions Trend (Top 5 Countries, 2000–2022)
Country
Asia
Asia (GCP)
High-income countries
Non-OECD (GCP)
World
CO₂ Emissions
30B
20B
10B
2000
2005
2010
2015
2020
Year

In [10]:

```python
import pandas as pd
import plotly.express as px
import plotly.io as pio
import os

pio.renderers.default = 'browser'
print("Current directory:", os.getcwd())
df = pd.read_csv(r"C:\Users\SANDEEP MEDIDA\Desktop\data-viz-class-materia
df = df.rename(columns={
    'Entity': 'Country',
    'Annual CO₂ emissions': 'CO2'
})

print("\nColumns after rename:", df.columns)
df['CO2_Mt'] = df['CO2'] / 1_000_000
df = df[~df['Country'].isin(['World', 'Asia', 'Europe', 'Africa'])]
region_map = {
    'Germany': 'Western Europe',
    'France': 'Western Europe',
    'United Kingdom': 'Western Europe',

    'India': 'South Asia',
    'Pakistan': 'South Asia',
    'Bangladesh': 'South Asia',

    'China': 'East Asia',
    'Japan': 'East Asia',
    'South Korea': 'East Asia',

    'Nigeria': 'Sub-Saharan Africa',
    'Kenya': 'Sub-Saharan Africa',
    'Ethiopia': 'Sub-Saharan Africa',

    'Brazil': 'Latin America',
    'Mexico': 'Latin America',
    'Argentina': 'Latin America'
}

df['Region'] = df['Country'].map(region_map)
df['Region'] = df['Region'].fillna('Other')
print("\nSample data:")
print(df[['Country', 'Year', 'CO2_Mt', 'Region']].head())

print("\nColumns:", df.columns)
print("\nCountries:", df['Country'].nunique())

print("\nCO2 range:", df['CO2_Mt'].min(), "to", df['CO2_Mt'].max(), "Mt")

print("\nRegional averages (2022):")
print(
    df[df['Year'] == 2022]
    .groupby('Region')['CO2_Mt']
    .mean()
    .sort_values(ascending=False)
    .round(1)
)
df_recent = df[(df['Year'] >= 2000) & (df['Year'] <= 2022)]

top_countries = df_recent.groupby('Country')['CO2_Mt'].sum().nlargest(5).
df_top = df_recent[df_recent['Country'].isin(top_countries)]

```

```python
fig = px.line(
    df_top,
    x='Year',
    y='CO2_Mt',
    color='Country',
    title="Top 5 CO₂ Emitters Trend (2000-2022)"
)

fig.update_layout(
    plot_bgcolor='white',
    xaxis_title="Year",
    yaxis_title="CO₂ Emissions (Megatonnes)"
)

fig.show()
```

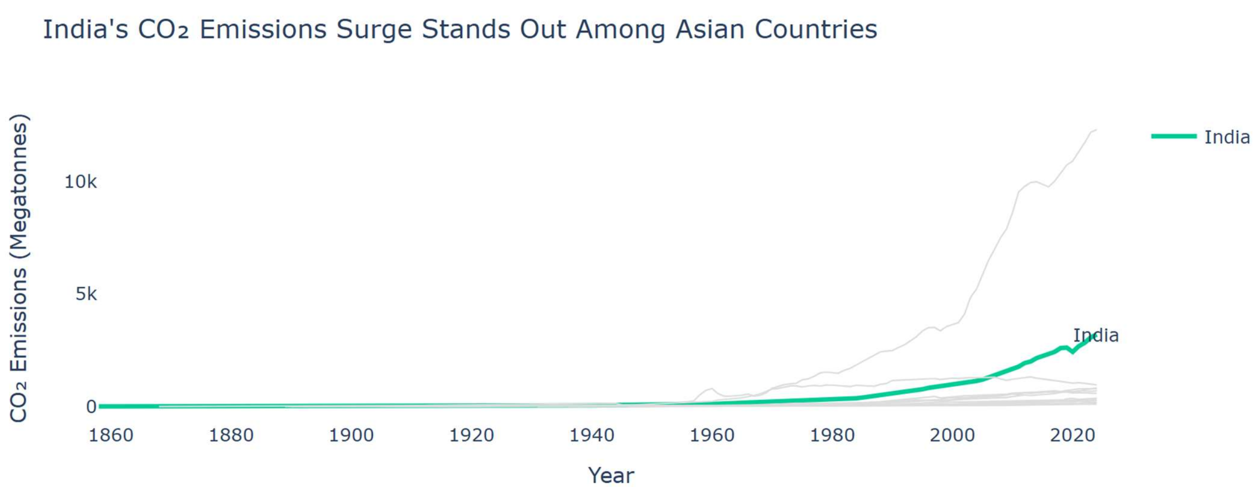
```
Current directory: C:\Users\SANDEEP MEDIDA

Columns after rename: Index(['Country', 'Code', 'Year', 'CO2'], dtype='objec
t')

Sample data:
       Country  Year    CO2_Mt Region
0  Afghanistan  1949  0.014656  Other
1  Afghanistan  1950  0.084272  Other
2  Afghanistan  1951  0.091600  Other
3  Afghanistan  1952  0.091600  Other
4  Afghanistan  1953  0.106256  Other

Columns: Index(['Country', 'Code', 'Year', 'CO2', 'CO2_Mt', 'Region'], dtype
='object')

Countries: 243

CO2 range: 0.0 to 26233.051 Mt

Regional averages (2022):
Region
East Asia             4449.4
South Asia            1048.4
Other                  679.8
Western Europe         424.8
Latin America          372.5
Sub-Saharan Africa      56.3
Name: CO2_Mt, dtype: float64
```



In [12]:

```
import pandas as pd
import plotly.graph_objects as go
import plotly.io as pio
pio.renderers.default = 'browser'
df = pd.read_csv(r"C:\Users\SANDEEP MEDIDA\Desktop\data-viz-class-materia
df = df.rename(columns={
   'Entity': 'Country',
    'Annual CO₂ emissions': 'CO2'
    })
    df['CO2_Mt'] = df['CO2'] / 1_000_000
    asia_countries = [
   'China', 'India', 'Japan', 'South Korea',
    'Indonesia', 'Saudi Arabia', 'Iran',
    'Pakistan', 'Bangladesh', 'Vietnam',
'Thailand', 'Malaysia', 'Philippines'
]
df_asia = df[df['Country'].isin(asia_countries)]
highlight_country = 'India'
fig = go.Figure()

    for country in df_asia['Country'].unique():
    country_df = df_asia[df_asia['Country'] == country]

    if country == highlight_country:
    fig.add_trace(go.Scatter(
    x=country_df['Year'],
          y=country_df['CO2_Mt'],
          mode='lines',
            name=country,
            line=dict(width=3)   # thicker
          ))
    else:
      fig.add_trace(go.Scatter(
x=country_df['Year'],
          y=country_df['CO2_Mt'],
```

```
            mode='lines',
              line=dict(color='#DDDDDD', width=1),
              showlegend=False
              ))
              highlight_df = df_asia[df_asia['Country'] ==
              highlight_country]

    fig.add_annotation(
    x=highlight_df['Year'].max(),
    y=highlight_df['CO2_Mt'].iloc[-1],
    text=highlight_country,
    showarrow=False,
    font=dict(size=12)
    )
    fig.update_layout(
    title=f"{highlight_country}'s CO₂ Emissions Surge Stands Out Among As
    xaxis_title="Year",
  yaxis_title="CO₂ Emissions (Megatonnes)",
    plot_bgcolor='white'
    )
    fig.show()
```



In [3]:

```python
import pandas as pd
import plotly.graph_objects as go
import plotly.io as pio
pio.renderers.default = 'browser'
df = pd.read_csv(r"C:\Users\SANDEEP MEDIDA\Desktop\data-viz-class-materia
df = df.rename(columns={
    'Entity': 'Country',
    'Annual CO₂ emissions': 'CO2'
})
df['CO2_Mt'] = df['CO2'] / 1_000_000
df = df[~df['Country'].isin(['World', 'Asia', 'Europe', 'Africa'])]
df['Region'] = 'Other'
grouped = df.groupby(['Region', 'Year'])['CO2_Mt'].mean().reset_index()
df_slope = grouped[grouped['Year'].isin([2000, 2022])]
pivot = df_slope.pivot(index='Region', columns='Year', values='CO2_Mt').d
fig = go.Figure()
for region in pivot.index:
    y0 = pivot.loc[region, 2000]
    y1 = pivot.loc[region, 2022]
    color = 'green' if y1 > y0 else 'red'

    fig.add_trace(go.Scatter(
        x=[2000, 2022],
        y=[y0, y1],
        mode='lines+markers+text',
        text=[round(y0,1), round(y1,1)],
        textposition=["middle left", "middle right"],
        line=dict(color=color, width=3),
        showlegend=False
    ))
fig.update_layout(
    title="CO₂ Change (2000 vs 2022)",
    yaxis=dict(showticklabels=False),
    plot_bgcolor='white'
)
fig.show()
```

In [ ]:

```python

```